

1. A function $f(x)$ defined on the interval $[-1, 1]$ has an antiderivative of $F(x)$. Assume that $F(-1) = 8$ and $F(1) = 7$. Which of the following statements below must be true? (circle all that apply)

(A) $\int_{-1}^1 f(x) dx = 1$
(B) $F(x)$ is an increasing function.
(C) $F(x)$ is an odd function.
(D) $\int_{-1}^1 f(x) dx = 0$
(E) $\int_{-1}^1 f(x) dx = -1$
☒ (F) None of the above.

We don't know if $f(x)$ is continuous on $[-1, 1]$ so we can't say that (A), (D), & (E) are correct.

$F(x)$ cannot be odd since we would need $F(-1) = -F(1)$, which is not true.

$F(x)$ would be increasing on $[-1, 1]$ if $f(x) > 0$ on $[-1, 1]$, but we don't know anything about the function $f(x)$ on this interval.

MAYBE TOO HARD FOR AN EXAM???? Good 5.3.1 honors question?

ALSO FOR HONORS

2. Find the interval $[a, b]$ for which the value of $\int_a^b (2 + x - x^2) dx$ is a maximum.

Maybe modify so that it can be approached graphically as well as using optimization.

3. If $f(x) = \int_0^{g(x)} \frac{1}{\sqrt{1+t^3}} dt$, where $g(x) = \int_0^{\cos(x)} [1 + \sin(t^2)] dt$, find $f'\left(\frac{\pi}{2}\right)$.

4. Find the maximum value of the area of the region under the curve $y = 4x - x^3$ from $x = a$ to $x = a + 1$, for all $a > 0$.